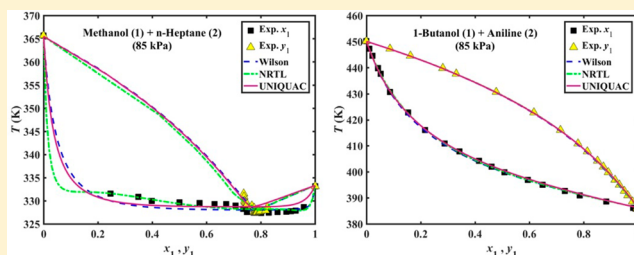


Experimental Vapor–Liquid Equilibria and Thermodynamic Modeling of the Methanol + *n*-Heptane and 1-Butanol + Aniline Binary Systems

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ABSTRACT: Isobaric vapor–liquid equilibria (VLE) of the binary systems of methanol + *n*-heptane and 1-butanol + aniline were measured at 85 kPa. The experiments were carried out using an Othmer still with the objective of measuring the equilibrium temperature, as well as the composition of the vapor and liquid phases as determined by gas chromatography analysis. According to the experimental results obtained, both of the binary mixtures studied demonstrate positive deviation from ideal behavior. Furthermore, a point of minimum-boiling azeotrope was observed for the methanol + *n*-heptane system. The VLE data measured have been verified to be thermodynamically consistent based on the Wisniak modification of the Herington area test for isobaric VLE data. Additionally, the two systems were thermodynamically modeled. Because of the relatively low pressure involved in this work, the modified Raoult's law was utilized for this purpose. In this regard, the experimental vapor–liquid equilibrium data were correlated with different excess Gibbs energy models, including the Wilson, the nonrandom two-liquid (NRTL), and the universal quasi-chemical (UNIQUAC) activity coefficient models. The results obtained show reasonably good agreement with the experimental data.



1. INTRODUCTION

Hydrocarbon–alcohol systems are among the interesting mixtures in the oil and gas industries, with wide-reaching applications. For example, alcohols are commonly used as hydrate inhibitors in gas pipelines,¹ utilized to improve the antiknock properties of gasoline, or used as alternative fuel components.²

Among different alkanols currently used in the petrochemical industries, methanol has significant importance because of its relatively inexpensive production cost.³ To design and optimize separation process equipment that involves mixtures of methanol + *n*-alkanes, accurate knowledge of the phase behavior of these highly complex systems is of great importance. As the thermodynamic behavior of these mixtures is strongly affected by the association interactions between the molecules, reliable knowledge of their experimental vapor–liquid equilibrium (VLE) is essential for the development of accurate predictive thermodynamic models, such as different equations of state or various activity coefficient models. Particularly, knowledge of the complex phase behavior of those systems that contain both strongly associating and nonassociating components is of especial interest, as such nonideal equilibria tend to exhibit interesting thermodynamic phenomena such as azeotropes and regions of vapor–liquid–liquid equilibria (VLLE).¹ Moreover, as the hydrocarbon mixtures handled in the petrochemical industries are in fact highly complex mixtures comprising a large number of individual hydrocarbon groups, considering all of the species involved would be a significantly cumbersome task.² In this regard, a much more tractable approach would be to study model systems of heavy

hydrocarbons + methanol, for instance the system of interest in the current work, methanol + *n*-heptane. A number of isothermal VLE data have been reported in the literature for the methanol + *n*-heptane binary system. Isobaric VLE data of this system are very limited as most of the studies present data that are isothermal and limited to only one temperature: Hongo et al.⁴ measured the VLE of methanol + hexane, methanol + heptane, ethanol + hexane, ethanol + heptane, and ethanol + octane by a flow-type apparatus at 298.15 (K). Kammerer et al.⁵ reported the thermodynamic excess properties and VLE data of several binary and ternary mixtures containing methanol, tert-amyl methyl ether, and an alkane by an analytical method at 313.15 (K). Oh and Park⁶ measured the VLE data for the ternary systems of methyl *tert*-butyl ether + methanol + methylcyclohexane and methyl *tert*-butyl ether + methanol + *n*-heptane and their corresponding binary systems by using headspace gas chromatography (HSGC) at 313.15 (K). Isobaric VLE data for the methanol + *n*-heptane system have only been reported at atmospheric pressure (Benedict et al.,⁷ Budantseva et al.,⁸ and Hirata et al.⁹), and at 54.13 (kPa) (by Zieborak and Maczynska¹⁰).

Some attempts have also been made to model this challenging system. Browarzik¹¹ performed phase-equilibrium

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calculations on the *n*-alkane + alkanol systems using continuous thermodynamics. The results obtained for the methanol + *n*-heptane system were compared with the experimental data of Hongo et al.⁴ A systematic study on methanol + *n*-alkane vapor–liquid and liquid–liquid equilibria was undertaken by Yarrison and Chapman¹ using the CK-SAFT¹² and PC-SAFT¹³ equations of state and the results were compared to the experimental data of Kammerer et al.⁵

Regarding the second system investigated, aniline finds widespread usage in the manufacture of rubber, dyes and pigments, polymers, agricultural chemicals, pharmaceuticals, and also photographic chemicals.¹⁴ In this regard, knowledge of the phase behavior of mixtures of aniline is important for a better understanding of the compositional behavior of these systems for accurate design of the separation processes of interest in industrial applications. Especially, detailed phase behavior of the complex aniline + alkanol mixtures which entail strong hydrogen bonds is necessary for the design of *N*-alkylation processes involving aniline.^{14,15} However, no experimental studies of the VLE phase behavior for the 1-butanol + aniline mixture have been reported as of yet.

In this study, isobaric VLE data of the binary systems of methanol + *n*-heptane, as well as 1-butanol + aniline, were measured at 85 kPa using the modified Othmer still. The experimental activity coefficients were determined and compared with those correlated by the activity coefficient models of Wilson,¹⁶ non-random two-liquid (NRTL),¹⁷ and universal quasi-chemical (UNIQUAC).^{18,19} For this purpose, the binary interaction parameters of these models were optimized using the genetic algorithm²⁰ through a program routine that includes bubble-point temperature calculations.²¹

2. MATERIALS AND METHODS

2.1. Materials. Methanol, with reagent grade purity (at least 99.8%), was purchased from Merck, and *n*-heptane, 1-butanol, and also aniline were obtained from Scharlua with an ultrapure grade (at least 99.5%), as detailed in Table 1. All of the chemicals were used without further purification.

Table 1. Specification of the Experimental Reagents

chemical name	CASRN	supplier	purity (mass percent) ^a	purification method
methanol	67-56-1	Merck	≥99.8%	none
<i>n</i> -heptane	142-82-5	Scharlua	≥99.5%	none
1-butanol	71-36-3	Scharlua	≥99.5%	none
aniline	62-53-3	Scharlua	≥99.5%	none

^aDeclared by the supplier.

2.2. Apparatus and Procedures. The modified Othmer still was used to carry out the isobaric VLE experiments. This apparatus consists of three main parts: the boiling flask, the condenser, and the condensate receiver.²² It operates dynamically by recirculating the vapor phase; that is, the vapor phase arising from the liquid phase is condensed within the condenser and collected in the condensate receiver. When the condensate receiver is completely filled, the condensate overflow is returned to the boiling flask. Recirculation continues until the equilibrium is reached (when the temperature does not change for 20 min). A SamWon ENG Co. PT100 (SU-105) precision thermometer with an accuracy of ±0.5 (K) was used to measure the equilibrium temperature. Eventually, when equilibrium is reached, samples of both the liquid and vapor phases are taken for analysis to determine their composition in two replicates. In this study, the samples were analyzed by a Varian 3700 GC flame

ionization detector (FID), using a (2 m × 1/8") 15% FFAP Chromosorb W-AW 80/100 packed column with an accuracy of ±0.03 in determining the molar compositions. The carrier gas used was helium, and also, the injection temperature, initial column temperature, temperature ramp rate, final column temperature, detector temperature, and the carrier gas flow rate were 493.15 (K), 363.15 (K), 8 (K·min⁻¹), 473.15 (K), 523.15 (K), 30 (mL·min⁻¹), respectively. Readers are referred to the work of Li et al.²³ for detailed information regarding the experimental apparatus and procedure. The ambient pressure was also measured using a Lambrecht KG Gottingen barometer with an accuracy of ±0.1 kPa. It should be noted that the apparatus employed in the current study, that is, the modified Othmer still, operates at atmospheric pressure. This means that the VLE data measured are isobaric at a constant pressure of 85 kPa, the ambient pressure in Shiraz, Iran.

2.3. Thermodynamic Consistency Test. To verify the reliability of any new experimental equilibrium data, it is a common practice to utilize the Gibbs–Duhem equation as a consistency test,²⁴ which interrelates the activity coefficients of all the components in a mixture. In this manner, if all of the necessary activity coefficients are available, the quality of any VLE data set could be verified by checking its compliance with the Gibbs–Duhem equation. Among the different integral and differential (slope) tests proposed thus far that employ the Gibbs–Duhem equation, the Herington²⁵ consistency test is the most widely used.²⁴ Incorporating the enthalpy of mixing, Wisniak²⁶ proposed a simple method for estimating the Herington criterion for isobaric VLE data, whereby the data are considered thermodynamically consistent if the condition of (*D* – *J*) being less than a value of 10 is met; where *D* and *J* are defined as in eqs 1 and (2):^{27,28}

$$D = 100 \frac{\left| \int_0^1 \ln \left(\frac{\gamma_1}{\gamma_2} \right) dx_1 \right|}{\int_0^1 \left| \ln \left(\frac{\gamma_1}{\gamma_2} \right) \right| dx_1} \quad (1)$$

$$J = 150 \frac{T_{\text{Max}} - T_{\text{Min}}}{T_{\text{Min}}} \quad (2)$$

where *T*_{Max} and *T*_{Min} are the maximum and minimum boiling temperatures in the system under study, respectively.

2.4. Thermodynamic Modeling. The basic relation concerning the equilibrium distribution of any component between the vapor and liquid phases can be directly derived from the criterion of the equality of chemical potentials of the components distributed in the two contacting phases. This can be expressed as

$$y_i P \hat{\phi}_i = x_i \gamma_i \phi_i^{\text{sat}} P_i^{\text{sat}} \exp \left[\frac{v_i^{\text{L}} (P - P_i^{\text{sat}})}{RT} \right] \quad (3)$$

where *y*_{*i*}, *x*_{*i*}, *γ*_{*i*}, *v*_{*i*}^L, and *P*_{*i*}^{sat} are the vapor phase mole fraction, liquid phase mole fraction, activity coefficient, liquid molar volume, and vapor pressure of component *i*, respectively, and *P* is the total pressure of the system. Moreover, *φ*_{*i*} and *φ*_{*i*}^{sat} represent the vapor phase fugacity coefficient at the system pressure and the pure component fugacity coefficient of component *i* at its saturation pressure, respectively.

Because of the relatively low pressure involved in the current work, the modified Raoult's law²¹ was adopted, whereby the vapor phase is assumed to be an ideal gas (hence the component

fugacity coefficients are assumed equal to unity), and the exponential term of eq 3, commonly known as the Poynting factor, is approximated as 1. Consequently, the experimental activity coefficients are evaluated using the following simplified equation:

$$\gamma_i = \frac{y_i P}{x_i P_i^{\text{sat}}} \quad (4)$$

The saturation pressure of the pure components of interest can be estimated using the Antoine equation, eq 5:

$$\log_{10}(P_i^{\text{sat}}/\text{bar}) = A_i - \frac{B_i}{T/K + C_i} \quad (5)$$

where the component-specific constants (A_i , B_i , C_i) of the components of interest were taken from the NIST database,²⁹ as listed in Table 2.

Table 2. Antoine Equation (eq 5) Constants of the Pure Components²⁹

component	A	B	C
methanol	5.158	1569.613	−34.846
<i>n</i> -heptane	4.028	1268.636	−56.199
1-butanol	4.390	1254.502	−105.246
aniline	4.345	1661.858	−74.048

By employing the genetic algorithm²⁰ as a robust optimization algorithm, coupled with a bubble-point temperature calculation routine,²¹ the binary interaction coefficients of the Wilson, NRTL, and UNIQUAC activity coefficient models were estimated. In doing so, the following objective function was used:

$$\text{O.F.} = \frac{1}{N} \sum_{i=1}^N \left(\frac{y_i^{\text{exp}} - y_i^{\text{calc}}}{y_i^{\text{exp}}} \right)^2 + \frac{1}{N} \sum_{i=1}^N \left(\frac{T_i^{\text{exp}} - T_i^{\text{calc}}}{T_i^{\text{exp}}} \right)^2 \quad (6)$$

where N is the number of data points, and the superscripts “exp” and “calc” denote the experimental and calculated values, respectively.

The UNIQUAC^{18,19} pure component parameters of r , q , and q' , which are molecular-structural constants and depend on the molecular size, external surface area, and the surface of interaction, respectively, were obtained from the literature²⁴ as given in Table 3. In keeping with the proposal of Anderson and

Table 3. Pure Component Structural Parameters of the UNIQUAC Activity Coefficient Model^{24,31}

component	r	q	q'
methanol	1.43	1.43	0.96
<i>n</i> -heptane	5.17	4.40	4.40
1-butanol	3.92	3.67	0.88
aniline	3.72	2.83	2.83

Prausnitz,³⁰ to improve the accuracy of the model for systems containing the lower alcohols, the surface of interaction parameter (q') of methanol, and also 1-butanol, used in the residual contribution term of the UNIQUAC model is taken to be smaller than its geometric external surface (q), which implies that the intermolecular attraction is dominated by the hydrogen bonding.²⁴ As can be seen in Table 3, for both *n*-heptane and aniline, $q = q'$.

3. RESULTS AND DISCUSSION

Following the experimental procedure outlined in section 2, the VLE data of interest were measured for the methanol (1) +

n-heptane (2), and the 1-butanol (1) + aniline (2) binary systems at 85 kPa using the refined Othmer apparatus. Table 4

Table 4. Experimental VLE Data for the Methanol (1) + *n*-Heptane (2) System at $P = 85$ kPa^a

T (K)	x_1	y_1	γ_1	γ_2
365.65	0	0		1
331.55	0.247	0.737	3.818	1.339
330.85	0.349	0.738	2.786	1.584
329.95	0.399	0.740	2.535	1.763
329.65	0.501	0.743	2.049	2.131
329.55	0.574	0.747	1.806	2.464
329.25	0.629	0.749	1.673	2.843
329.25	0.672	0.750	1.568	3.200
328.95	0.707	0.767	1.544	3.376
328.25	0.735	0.775	1.546	3.703
327.55	0.749	0.775	1.562	4.020
328.45	0.758	0.775	1.487	4.019
327.45	0.774	0.777	1.521	4.454
327.45	0.778	0.779	1.518	4.486
327.35	0.802	0.781	1.484	4.984
327.45	0.830	0.783	1.430	5.750
327.55	0.861	0.786	1.378	6.926
327.65	0.894	0.795	1.336	8.687
327.75	0.911	0.800	1.314	10.049
327.85	0.929	0.818	1.312	11.348
328.65	0.958	0.824	1.239	18.216
333.15	1	1	1	

^aThe expanded uncertainties U for a 0.95 level of confidence are $U(T) = 0.5$ K, $U(P) = 0.1$ kPa, and $U(x) = U(y) = 0.03$.

and Table 5 list the experimental data obtained including the temperature, and the molar compositions of the vapor and

Table 5. Experimental VLE Data for the 1-Butanol (1) + Aniline (2) System at $P = 85$ kPa^a

T (K)	x_1	y_1	γ_1	γ_2
450.25	0	0		1
447.25	0.010	0.087	1.390	0.996
444.65	0.022	0.161	1.378	0.996
439.85	0.043	0.281	1.355	0.996
437.75	0.052	0.330	1.345	0.996
430.65	0.087	0.479	1.307	0.997
422.85	0.153	0.617	1.259	1.000
416.00	0.216	0.715	1.212	1.006
410.95	0.288	0.779	1.173	1.015
407.85	0.342	0.814	1.148	1.023
404.25	0.415	0.851	1.118	1.038
402.00	0.461	0.872	1.100	1.051
399.95	0.508	0.891	1.084	1.066
397.00	0.594	0.916	1.063	1.094
395.00	0.649	0.932	1.050	1.119
392.65	0.731	0.950	1.036	1.155
390.95	0.785	0.963	1.029	1.186
388.65	0.883	0.980	1.022	1.238
386.25	0.982	0.997	1.019	1.304
386.00	0.991	0.998	1.019	1.311
385.75	1	1	1	

^aThe expanded uncertainties U for a 0.95 level of confidence are $U(T) = 0.5$ K, $U(P) = 0.1$ kPa, and $U(x) = U(y) = 0.03$.

liquid phases, alongside the corresponding activity coefficients at each temperature.

Table 6. Thermodynamic Consistency Test Results (eqs 1 and 2)

system	<i>D</i>	<i>J</i>	<i>D</i> − <i>J</i>
methanol + <i>n</i> -heptane	1.59	17.55	−15.96
1-butanol + aniline	19.39	25.08	−5.69

To be able to perform the required integration on the experimental data in the Wisniak modification of the Herington area consistency test detailed in section 2.3, a linear line was fitted to the corresponding $\ln(\gamma_1/\gamma_2)$ versus x_1 data. The results, as well as a graphical illustration of this test, can be seen in Table 6 and Figure 1, respectively. The values of (*D* − *J*) were found to be less than 10 for both of the systems under consideration, usually taken as the criterion of acceptable quality and thermodynamic consistency of experimental isobaric VLE data.³²

The interaction parameters of the three activity coefficient models studied for both of the systems of interest, estimated following the approach detailed in section 2.4, are reported in Table 7. Moreover, the relative mean squared error (RMSE) relation is used in Table 8 to compare the experimental and correlated values.

The temperature–composition (*T*–*x*,*y*) diagram for the experimental data, as well as the values obtained by bubble-point temperature calculations employing the three activity coefficient models, can be seen in Figure 2. It should be noted that the few experimental isobaric data pertaining to the methanol + *n*-heptane system previously reported in the literature have also been included in this figure. Furthermore, an illustration comparing the estimated and the correlated activity coefficients is depicted in Figure 3. The methanol + *n*-heptane binary system shows a minimum-boiling azeotropic point at 85 kPa; as can be seen in Figure 2, the same phenomenon could also be observed at 101.33 kPa,⁸ and most likely, at 54.129 kPa¹⁰ as well. As can be

inferred from Figure 2, as well as the RMSE values reported in Table 8, the equilibrium temperatures calculated using the NRTL model have a higher accuracy as compared with those obtained using the Wilson and UNIQUAC models. Nonetheless, the calculated vapor phase mole fractions using the UNIQUAC model demonstrate lower deviations compared with those estimated using the NRTL and Wilson models. All in all, considering the average value of the deviations of both the vapor phase mole fractions and the equilibrium temperatures, the UNIQUAC model manifests a slightly better performance. As for the 1-butanol + aniline system and based on the RMSE deviations listed in Table 8, the NRTL model

Table 8. RMSE Deviations of the Correlated Values^a

system	NRTL ¹⁷		Wilson ¹⁶		UNIQUAC ^{18,19}	
	$\Delta y \times 10^3$	$\Delta T \times 10^3$	$\Delta y \times 10^3$	$\Delta T \times 10^3$	$\Delta y \times 10^3$	$\Delta T \times 10^3$
methanol (1) + <i>n</i> -heptane (2)	1.037	0.098	0.553	0.263	0.448	0.296
1-butanol (1) + aniline (2)	0.281	0.000	0.675	0.001	0.532	0.001

^aCalculations:

$$\Delta = \frac{1}{N} \sum_{i=1}^N \left(\frac{\text{Exp. Value}_i - \text{Calc. Value}_i}{\text{Exp. Value}_i} \right)^2$$

demonstrates a better correlative ability for the equilibrium temperatures and compositions. However, in regards to the correlation of the experimental activity coefficients, as can be seen in Figure 3, the UNIQUAC model is more accurate in an extended composition range, while the NRTL model is accurate mainly in the intermediate composition range.

Moreover, Figure 4 gives a plot of vapor phase mole fractions versus liquid phase mole fractions for both the experimental

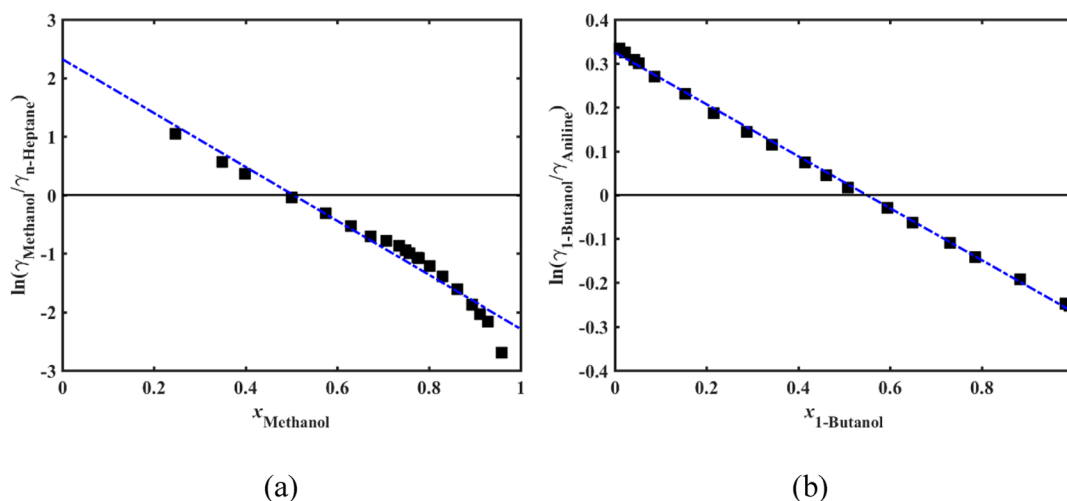


Figure 1. Herington consistency test diagram for the experimental VLE data (■, experimental data; ---, fitted line): (a) methanol + *n*-heptane ($R^2 = 0.957$); (b) 1-butanol + aniline ($R^2 = 0.998$).

Table 7. Estimated Binary Interaction Parameters of the Activity Coefficient Models for the Methanol + *n*-Heptane and 1-Butanol + Aniline Systems^a

system	NRTL ¹⁷			Wilson ¹⁶		UNIQUAC ^{18,19}	
	<i>A</i> ₁₂	<i>A</i> ₂₁	α	<i>A</i> ₁₂	<i>A</i> ₂₁	<i>A</i> ₁₂	<i>A</i> ₂₁
methanol (1) + <i>n</i> -heptane (2)	1099.503	932.945	0.480	871.050	788.550	5.822	950.947
1-butanol (1) + aniline (2)	133.830	1.630	0.192	154.920	0.490	−49.340	40.450

^aCalculations: $A_{12} = \frac{\lambda_{12} - \lambda_{11}}{R}$; $A_{21} = \frac{\lambda_{21} - \lambda_{22}}{R}$; where *R* is the universal gas constant.

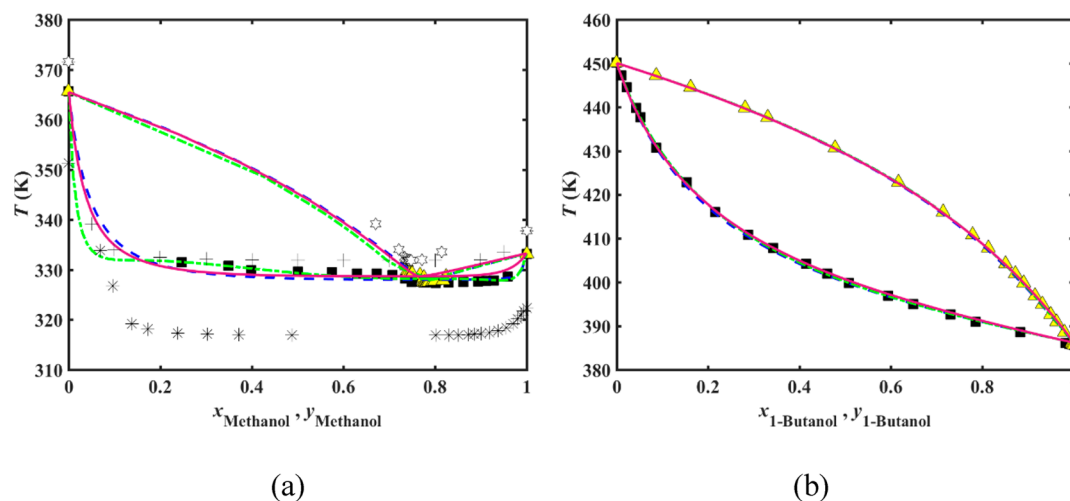


Figure 2. Temperature–composition diagram: ■, experimental liquid phase mole fraction (this study); yellow ▲, experimental vapor phase mole fraction (this study); green ---, correlated with the NRTL model;¹⁷ blue ---, correlated with the Wilson model;¹⁶ pink —, correlated with the UNIQUAC model.^{18,19} (a) Methanol + *n*-heptane, $P = 54.129$ (kPa):¹⁰ (*) exp x ; $P = 101.330$ (kPa):⁸ (+) exp. x , (☆) exp. y ; (b) 1-butanol + aniline.

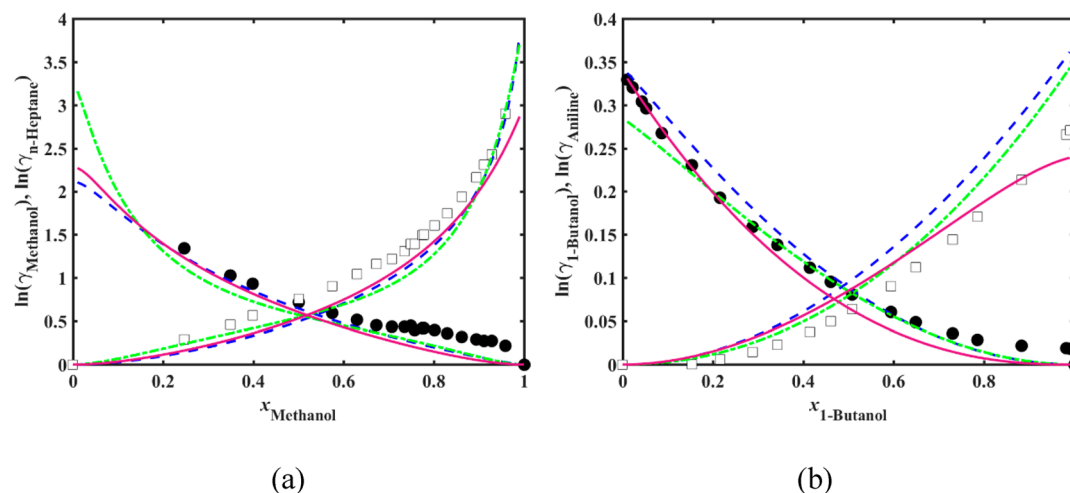


Figure 3. Experimental data as well as the correlated values of the activity coefficients (γ_i): ●, experimental γ_{methanol} , $\gamma_{1\text{-butanol}}$; ■, experimental $\gamma_{n\text{-heptane}}$, γ_{aniline} ; green ---, correlated with the NRTL model;¹⁷ blue ---, correlated with the Wilson model;¹⁶ pink —, correlated with the UNIQUAC model.^{18,19} (a) Methanol + *n*-heptane; (b) 1-butanol + aniline.

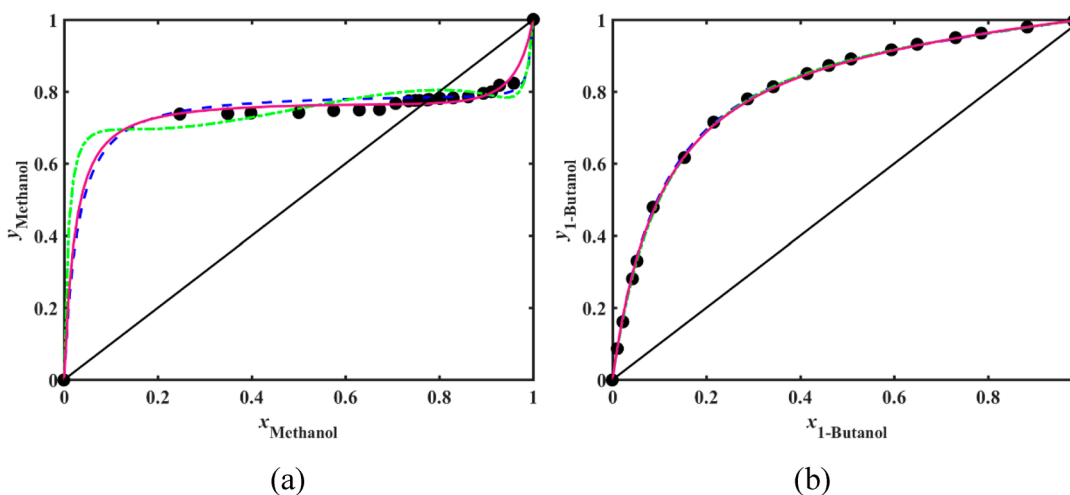


Figure 4. Vapor phase mole fraction versus the liquid phase mole fraction: ●, experimental data; green ---, correlated with the NRTL model;¹⁷ blue ---, correlated with the Wilson model;¹⁶ pink —, correlated with the UNIQUAC model.^{18,19} (a) Methanol + *n*-heptane; (b) 1-butanol + aniline.

data and the calculated results with the three activity models. These two figures show that both of the systems of interest in this work have positive deviation from Raoult's law; however, a closer look at the experimental activity coefficients reported in Table 4 and Table 5 reveals that this deviation from ideality is less manifested in the 1-butanol + aniline system. It is worthwhile to point out that the NRTL model predicts a region of phase split for the methanol + *n*-heptane system, while such a phenomenon was not observed during the experiments. This is unlike the UNIQUAC and Wilson models, which accurately estimate the type of phase behavior of the methanol + *n*-heptane system.

4. CONCLUSIONS

In this study, isobaric VLE of the binary mixtures of methanol + *n*-heptane and 1-butanol + aniline were measured experimentally at 85 (kPa). Both of the systems studied show positive deviation from the ideal Raoult's law behavior, even to the extent of demonstrating a minimum-boiling azeotrope point in the methanol + *n*-heptane mixture. The quality of the experimental data was examined by a widely used thermodynamic consistency test for isobaric VLE data. In this regard, the Wisniak modification of the Herington area consistency test demonstrated the reasonable reliability of the reported data. Thermodynamic modeling of the systems of interest was also carried out whereby the Wilson, NRTL, and, UNIQUAC activity coefficient models were employed to correlate the equilibrium data using the modified Raoult's law. In doing so, the experimental activity coefficients were calculated, and the binary interaction parameters of the activity coefficient models were optimized and reported. The results indicate that the UNIQUAC model demonstrates a higher level of accuracy relative to the Wilson and NRTL models for the methanol + *n*-heptane system. The NRTL activity model even predicts a phase split which is not observed experimentally. Similarly, for the 1-butanol + aniline system the UNIQUAC model exhibits a more reliable correlation of the equilibrium properties in an extended composition range.

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Notes

The authors declare no competing financial interest.

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